Lab No: 11

✵ **Lab Title:** **Strings**

○ **Learning objectives:**

The following topics will be covered in this lab session

- What is a String
- When to use strings
- Declaration of strings
- Initialization of strings
- Accessing strings element

STRINGS:

It is defined as character array that is terminated by a null. A null character is either specified by using '\0' or zero. Two kinds of strings are commonly used in C++i.e. C-strings and strings .both are objects of the string class. C-strings are arrays of type char. These strings are known as *C-strings*, or *C-style strings*, because they were the only kind of strings available in the C language.

- **Example # 1:**

Write code for a program that asks the user to enter a string, and places this string in the string variable. Then it displays the string.

```
// stringin.cpp
// simple string variable
#include <iostream>
Using namespace std;
int main()
{
const int MAX = 80;   //max characters in string
char str[MAX];          //string variable str
cout << "Enter a string: ";
cin >> str; //put string in str
//display string from str
cout << "You entered: " << str << endl;
return 0;
}
```

⇘ Output:

```
Enter a string: Malakand
You entered: Malakand
```

- **Example # 2:**

There is no built-in mechanism in C++ to keep a program from inserting array elements outside an array. However, it is possible to tell the >> operator to limit the number of characters it places in an array. This Example demonstrates this approach.

```
// safetyin.cpp
// avoids buffer overflow with cin. width
#include <iostream>
#include <iomanip> //for setw
int main()
{
const int MAX = 20; //max characters in string
char str[MAX]; //string variable str
cout << "\nEnter a string: ";
cin >> setw(MAX) >> str; //put string in str,
// no more than MAX chars
cout << "You entered: " << str << endl;
return 0;
}
```

Lab Tasks

- **Task 1: String Length Calculation:**

    1. Write a C program that declares and initializes a string variable called "message" with a phrase of your choice.
    2. Use a built-in function to calculate and display the length of the string.

⇘ Input:

```
#include <iostream>      // input/output ke liye library
#include <cstring>      // strlen() function use karne ke liye
using namespace std;     // standard namespace use kar rahe hain

int main()
{
   char message[100];   // 100 characters ka string array banaya

   cout << "Enter a message: ";    // user ko message enter karne ko keh rahe hain
   cin.get(message, 100);         // full line input le raha hai (spaces ke sath)

   cout << "You entered: " << message << endl;   // input print kar raha hai

   cout << "Length of string: " << strlen(message) << endl;
   // strlen() string ki length nikalta hai (null '\0' ko ignore karta hai)

   Return  0;   // program successful end
}
```

Output:

```
Hello, World! Size :13
```

- **Task 2: String Concatenation**

    1. Declare two string variables called "first Name" and "last Name" and initialize them with your first and last name.
    2. Concatenate the two strings to form a full name and display it.

Input;

```
#include <iostream>     // input/output ke liye
#include <cstring>      // strcpy() aur strcat() ke liye
using namespace std;    // standard namespace

int main()
{
    char firstName[50];   // first name store karne ke liye array
    char lastName[50];    // last name store karne ke liye array
    char fullName[100];   // dono names combine karne ke liye array

    cout << "Enter first name: ";   // user se first name lena
    cin >> firstName;               // input lena (sirf single word)

    cout << "Enter last name: ";    // user se last name lena
    cin >> lastName;                // input lena

    strcpy(fullName, firstName);    // first name ko fullName mein copy karna

    strcat(fullName, " ");          // beech mein space add karna

    strcat(fullName, lastName);     // last name ko join karna

    cout << "Full Name: " << fullName << endl;
    // final combined name display karna

    return 0;   // program end
}
```

➪ **Output:**

Full Name: Ali Khan

Conclusion:

In this lab, we learned the basic concepts of strings in C++. We studied two types of strings: **C-style strings (character arrays)** and **C++ string objects**. We understood how strings are stored in memory and the importance of the null character `\0` in C-strings.

We also practiced different input methods such as `cin`, `cin.get()`, and `getline()` to read strings, including full sentences with spaces. In addition, we used important string handling functions like `strlen()`, `strcpy()`, and `strcat()` to calculate string length, copy strings, and concatenate (join) strings.